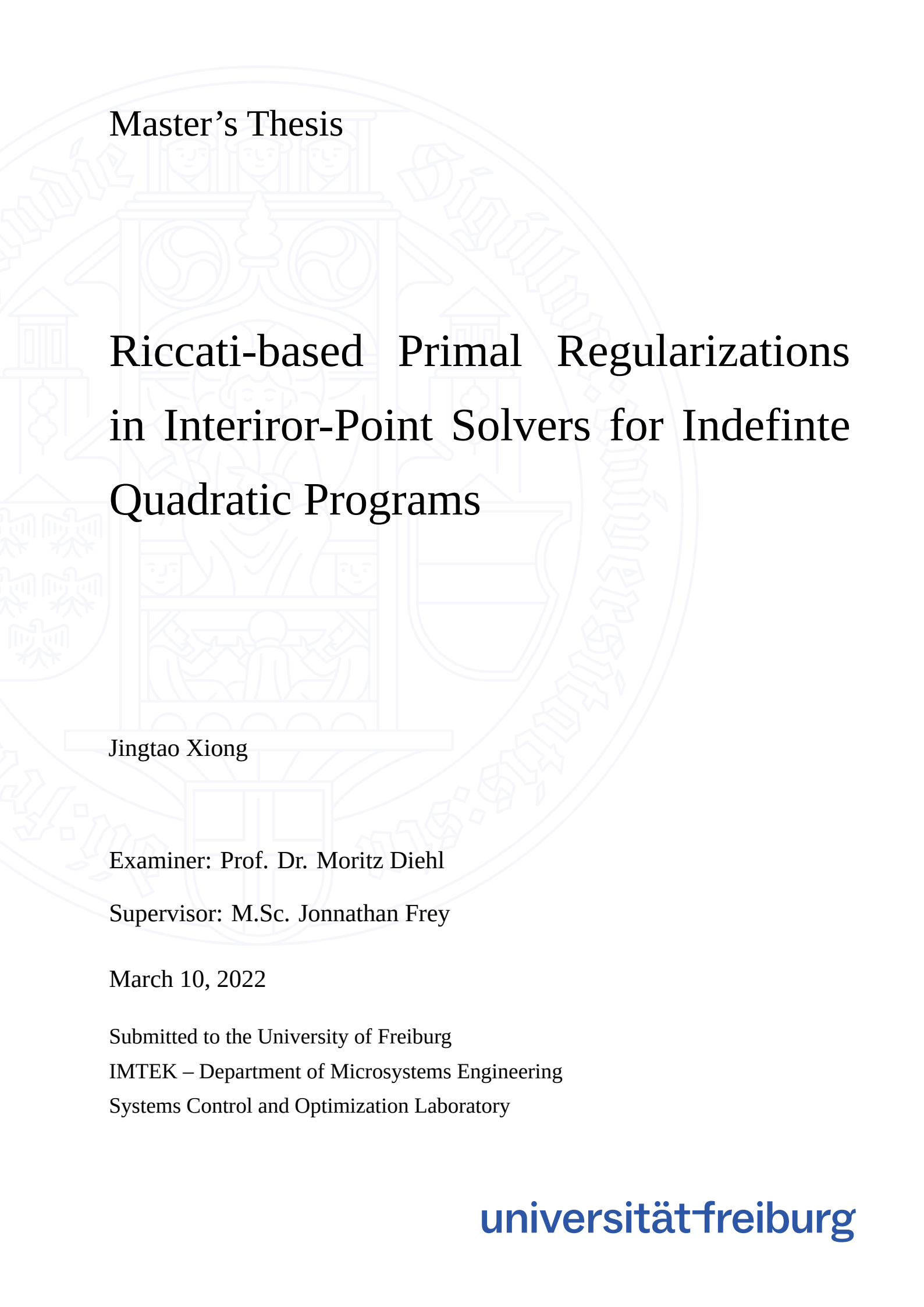




Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

Jingtao Xiong

Examiner: Prof. Dr. Moritz Diehl

Supervisor: M.Sc. Jonnathan Frey

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IMTEK – Department of Microsystems Engineering

Systems Control and Optimization Laboratory

University of Freiburg
IMTEK – Department of Microsystems Engineering
Systems Control and Optimization Laboratory

Author	Jingtao Xiong, Matriculation Number: 5986123
Editing Time	July 13, 2021 - March 10, 2022
Examiners	Prof. Dr. Moritz Diehl, IMTEK – Department of Microsystems Engineering Systems Control and Optimization Laboratory Prof. Dr. Max Doe, IMTEK – Department of Microsystems Engineering Laboratory for something that likely has a long title as well
Supervisor	M.Sc. Jonnathan Frey, IMTEK – Department of Microsystems Engineering Systems Control and Optimization Laboratory
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Chapter 1

A Benchmark for Indefinite OCP Quadratic Programs

In order to evaluate the performance of the proposed methods, especially in the context of nonlinear OCPs. We have created a benchmark consisting of a set of indefinite OCP quadratic programs from QP subproblems arising in the SQP method for solving nonlinear OCPs. The benchmark is designed to cover a wide range of problem instances, varying in size, structure, and degree of indefiniteness.

1.1 Benchmark Construction

1.1.1 Formulation of OCPs

Following [1], each system is cast as a fixed-time regulator problem. The eight OCPs are characterized as follows:

- **Pendulum:** An inverted pendulum on an actuated cart, driven by a horizontal force. It is regulated from the downward to the upright equilibrium subject to an input bound and an additional mixed state–input inequality constraint.
- **Acrobot:** A two-link pendulum actuated only at the elbow joint, with joint friction. The state is driven between two prescribed configurations through terminal equality constraints, yielding a strongly nonlinear, underactuated maneuver.
- **Dubins car:** A kinematic unicycle with position and heading as states, actuated by forward velocity and turn rate. It performs a parallel-parking maneuver, a lateral displacement returning to the initial heading, within a narrow corridor imposed by state-box constraints, with a terminal equality

constraint on the full pose.

- **Furuta pendulum:** A rotary inverted pendulum with experimentally identified parameters and joint damping, actuated by a single base-arm torque. The pendulum is swung up and stabilized at the upright position under tight torque limits and a non-uniform time grid.
- **Flying robot:** A planar free-flying robot with position, orientation and their rates, propelled by four non-negative thrusters that together produce a body-aligned net force and a turning moment. The full state is regulated to the origin under thrust non-negativity and summed-thrust linear constraints.
- **Robot arm:** A three-DOF robot arm (radial extension, rotation, elevation) with configuration-dependent inertia and a cross-axis actuator coupling that is nonlinear in the inputs. A joint angle is regulated to a new set-point under state and input bounds; the input coupling renders the exact Hessian indefinite.
- **Crane:** An overhead crane with hoisting length, trolley position and payload swing angle, actuated by trolley and hoisting accelerations. The load is transferred between two rest positions while respecting a swing-rate bound and a nonlinear region constraint on the payload.
- **Moonlander:** A planar lunar lander with translational and rotational motion under reduced gravity, actuated by two laterally offset thrusters. It performs a soft landing, reaching a ground target with zero terminal velocity, under non-negative, bounded thrust.

Table 1.1: Summary of the evaluated OCP problems

Problem	n_x	n_u	N
Pendulum	4	1	20
Acrobot	4	2	15
Dubins car	3	2	25
Furuta	4	1	20
Flying robot	6	4	30
robot arm	6	3	30
crane	6	2	35
moonlander	6	2	30

These OCPs also cover different state and control dimensions, as shown in Table 1.1. The state dimension n_x ranges from 3 to 6, the control dimension n_u from 1 to 4, and the horizon length N from 15 to 35. This spread is chosen so that the benchmark spans a range of common problem scales in control problems,

which allows the evaluated methods to be assessed across problems of varying size and structure. And several of the systems (e.g. the pendulum, acrobot and Furuta pendulum) are underactuated, which makes the regulation task more demanding and provides a useful source of difficulty in addition to the varying problem dimensions.

To be noticed, each problem is formulated as a regulator problem with a quadratic stage cost penalizing deviations from the target state and the control effort, subject to box constraints on the controls. This ensures the corresponding QP subproblems share the block-wise structure in the Hessian matrix, align with the targeted mathematical structure.

1.1.2 Collecting Strategy

In order to benchmark the proposed Riccati-based regularizations, we are interested in the QP subproblems with an indefinite reduced Hessian.

To obtain a representative set of such problems, we therefore extract the QP subproblems that arise inside the SQP process itself, rather than constructing them artificially. This keeps the collected QPs close to the dynamics and constraints encountered during an actual OCP solve.

Based on this idea, we implement a systematic approach with the following steps to create indefinite OCP QPs from the formulated OCP:

1. **Generate:** In this step, we generate iterates by solving the OCP. To obtain iterates at various points during the solution process, we can apply SQP-level regularization techniques (such as PROJECT or MIRROR) or utilize the Gauss-Newton Hessian approximation in `acados`.
2. **Collect:** In this step, we utilize the previously generated iterates as initial guesses for linearization within the solver. By restricting the maximum number of SQP iterations, we can selectively collect indefinite QPs that exhibit an indefinite reduced Hessian.

In the collect step, limiting the number of SQP iterations prevent the solver moving towards the neighborhood of the OCP solution, Thus the iterates are more likely to yield an indefinite linearization and are therefore the ones of interest here.

With this strategy, we are able to collect QP subproblems that will appear during the SQP solving process, which can provide relatively realistic dynamics and constraints in the QP problems. And the resulting QPs share the mathemat-

ical structure as:

$$\begin{aligned}
& \underset{x,u}{\text{minimize}} && \sum_{k=0}^{N-1} \frac{1}{2} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^T \begin{bmatrix} Q_k & S_k^T \\ S_k & R_k \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} q_k \\ r_k \end{bmatrix}^T \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \frac{1}{2} x_N^T Q_N x_N + q_N^T x_N \\
& \text{subject to} && x_0 = \bar{x}_0, \\
& && x_{k+1} = A_k x_k + B_k u_k, \quad k = 0, \dots, N-1, \\
& && \begin{bmatrix} \underline{u}_k \\ \underline{x}_k \\ \underline{d}_k \end{bmatrix} \leq \begin{bmatrix} J_k^{b,u} & 0 \\ 0 & J_k^{b,x} \\ D_k & C_k \end{bmatrix} \begin{bmatrix} u_k \\ x_k \end{bmatrix} \leq \begin{bmatrix} \bar{u}_k \\ \bar{x}_k \\ \bar{d}_k \end{bmatrix}, \quad k = 0, \dots, N-1
\end{aligned}$$

where, at each stage k , $x_k \in \mathbb{R}^{n_x}$ and $u_k \in \mathbb{R}^{n_u}$ denote the state and control (deviation) variables, and N is the horizon length. All remaining quantities are obtained by linearizing the OCP at the current SQP iterate:

- $Q_k = Q_k^T$, $R_k = R_k^T$ and S_k are the state, control, and state-control coupling blocks of the Hessian of the Lagrangian, while q_k and r_k are the corresponding gradient terms; Q_N and q_N are the terminal Hessian and gradient.
- $\bar{x}_0$ is the fixed initial state, and A_k , B_k are the Jacobians of the discretized dynamics with respect to x_k and u_k , which define the linearized state-transition equation.
- $J_k^{b,u}$ and $J_k^{b,x}$ are selection matrices picking out the bounded control and state components, with bounds $\underline{u}_k, \bar{u}_k$ and $\underline{x}_k, \bar{x}_k$ (box constraints).
- D_k and C_k are the Jacobians of the general path constraints with respect to u_k and x_k , bounded by $\underline{d}_k$ and $\bar{d}_k$; these constraints couple states and controls.

1.2 Benchmark Analysis

In this section, we will introduce some analysis of the collected problem instances for gaining a better understanding of characteristics regarding definiteness in the indefinite OCP QP benchmark. The analysis focuses on the definiteness of reduced Hessian matrices concerning how the indefiniteness of reduced Hessian distribute and where the indefiniteness results from.

1.2.1 Distribution of Indefiniteness

We first characterize the collected QPs along two axes: the *depth* of indefiniteness, measured by the magnitude of the smallest (most negative) eigenvalue $|\lambda_{\min}|$ of the reduced Hessian, and the *width*, measured by the fraction of block reduced Hessians that are indefinite within a QP (the indefinite-stage rate).

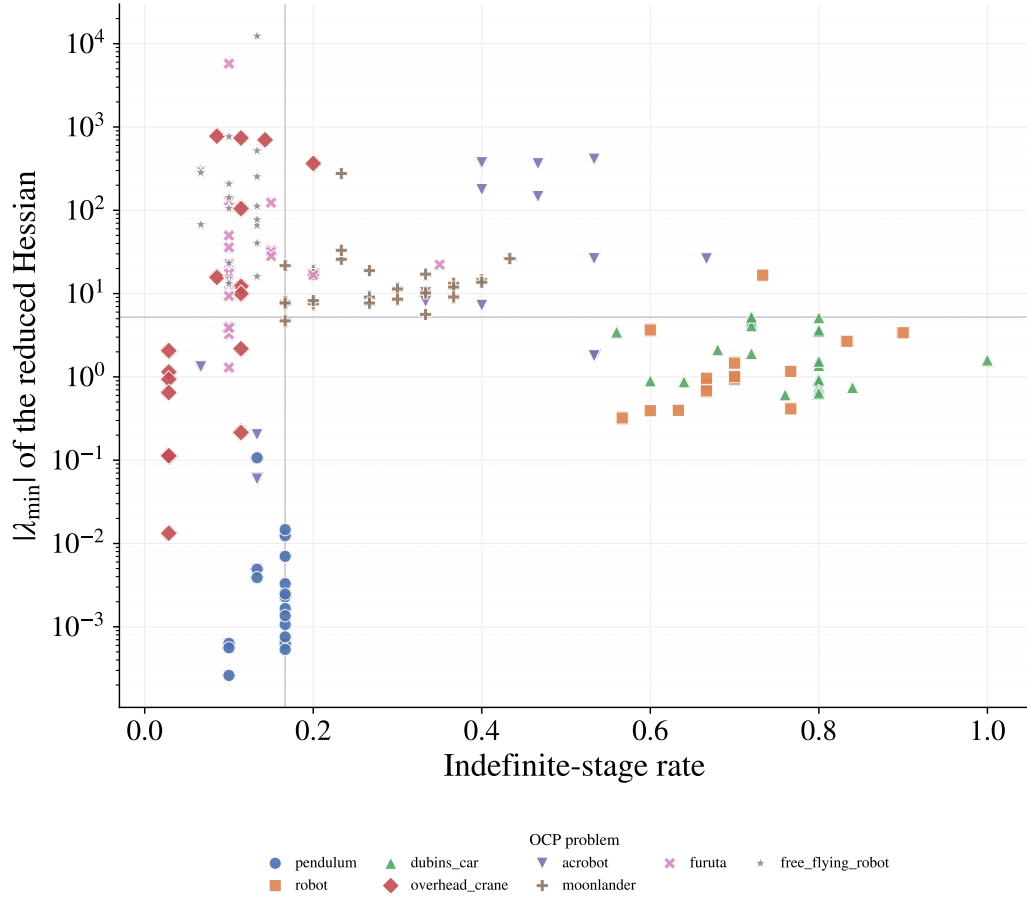


Figure 1.1: Depth ($|\lambda_{\min}|$ of the reduced Hessian, log scale) versus width (indefinite-stage rate) of the collected QPs, coloured by OCP problem.

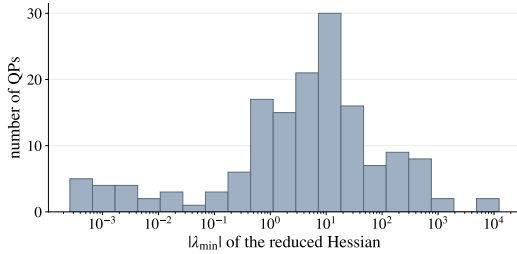
As seen from Figure 1.1, the collected indefinite QPs do not concentrate in a single regime. They spread over roughly eight orders of magnitude in depth ($|\lambda_{\min}|$ ranging from about 10^{-4} to 10^4). And the coverage is strongly problem-dependent: each OCP occupies a characteristic region of the depth–width plane rather than scattering uniformly.

Two complementary major patterns can be observed. Problems that are indefinite in only a small fraction of their stages, such as `overhead_crane`, `furuta`

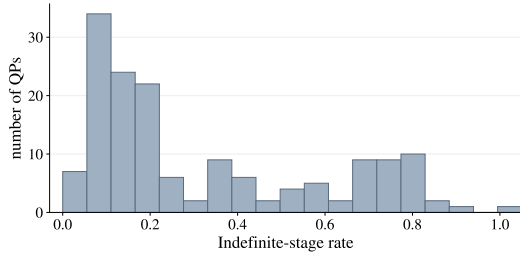
and `free_flying_robot`, nevertheless reach very deep negative curvature, with $|\lambda_{\min}|$ up to 10^2 – 10^4 . Conversely, problems with a high indefinite-stage rate, `dubins_car` and `robot`, with ranging from 0.6–1.0, tend to exhibit only moderate depth, with $|\lambda_{\min}|$ around 10^{-1} – 10^1 . Broad and deep indefiniteness thus appear to occur largely separately rather than together.

Some problems also fall in the middle, such as `acrobot` and `moonlander`, which occupy intermediate indefinite-stage rates (roughly 0.15–0.55) together with intermediate depths ($|\lambda_{\min}|$ mostly between 10^0 and 10^2 , occasionally deeper). They neither confine their negative curvature to a few stages nor spread it across the whole horizon, and thus bridge the two extreme regimes rather than belonging clearly to either.

Taken together, the eight OCPs cover a broad range of the depth–width plane: from narrow-but-deep (`overhead_crane`, `furuta`, `free_flying_robot`), through intermediate (`acrobot`, `moonlander`), to broad-but-shallow (`dubins_car`, `robot`), with `pendulum` adding cases of very shallow negative curvature at low indefinite-stage rates. This spread indicates that the collected QP subproblems exercise indefiniteness across qualitatively different regimes rather than a single one.



(a) Depth: distribution of $|\lambda_{\min}|$.



(b) Width: distribution of the indefinite-stage rate.

Figure 1.2: Marginal distributions of depth and width over all collected QPs.

Figure 1.2 shows the two marginal distributions of the same data. The depth histogram (Figure 1.2a) is centred around $|\lambda_{\min}| \approx 10$, where most of the indefinite QPs lie, and decays toward both a near-singular tail near 10^{-3} and a small number of very deep cases at 10^3 – 10^4 . Hence, although extreme cases exist, the negative curvature typically encountered during the SQP iterations is of moderate magnitude.

The width histogram (Figure 1.2b) shows how the indefinite-stage rate is distributed over the collected QPs. The rate covers the full range up to 1, with a

large share of QPs being indefinite in only a minority of their stages and a thinner spread toward higher rates.

This is consistent with the scatter plot: for most problems only a subset of stages become indefinite, whereas a few problems (`dubins_car`, `robot`) are indefinite across the majority of their stages.

1.2.2 Source of Indefiniteness

Recalling the block reduced Hessian at k stage is calculated as $M_k = R_k + B^\top P_{k+1} B$, which is assembled from two contributions: the control cost block R and the cost-to-go block P .

To understand *where* the indefiniteness originates, we classify every indefinite stage according to which of the two blocks has a negative smallest eigenvalue: *R only* (R indefinite, P definite), *P only* (P indefinite, R definite), and *RP only* (both indefinite).

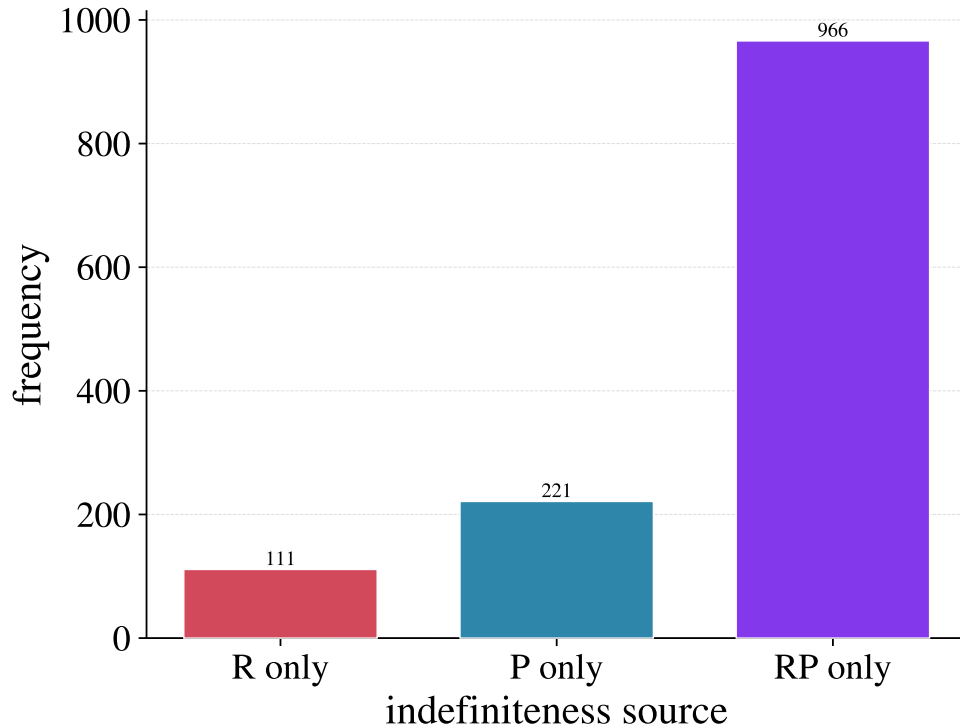


Figure 1.3: Number of indefinite stages by source: control cost R , cost-to-go P , or both.

Figure 1.3 aggregates all indefinite stages across the benchmark. The dominant source is the joint case: 966 of the 1298 indefinite stages have *both* R and P

indefinite, so in the large majority of cases the negative curvature is not confined to a single block. Among the remaining stages where only one block is indefinite, the cost-to-go block is the more frequent culprit: P alone accounts for 221 stages, roughly twice the 111 stages in which only the control cost R is indefinite.

Overall, the cost-to-go block P is indefinite more often than the control cost block R , which indicates that the indefiniteness is driven primarily by the cost-to-go term rather than by the control cost.

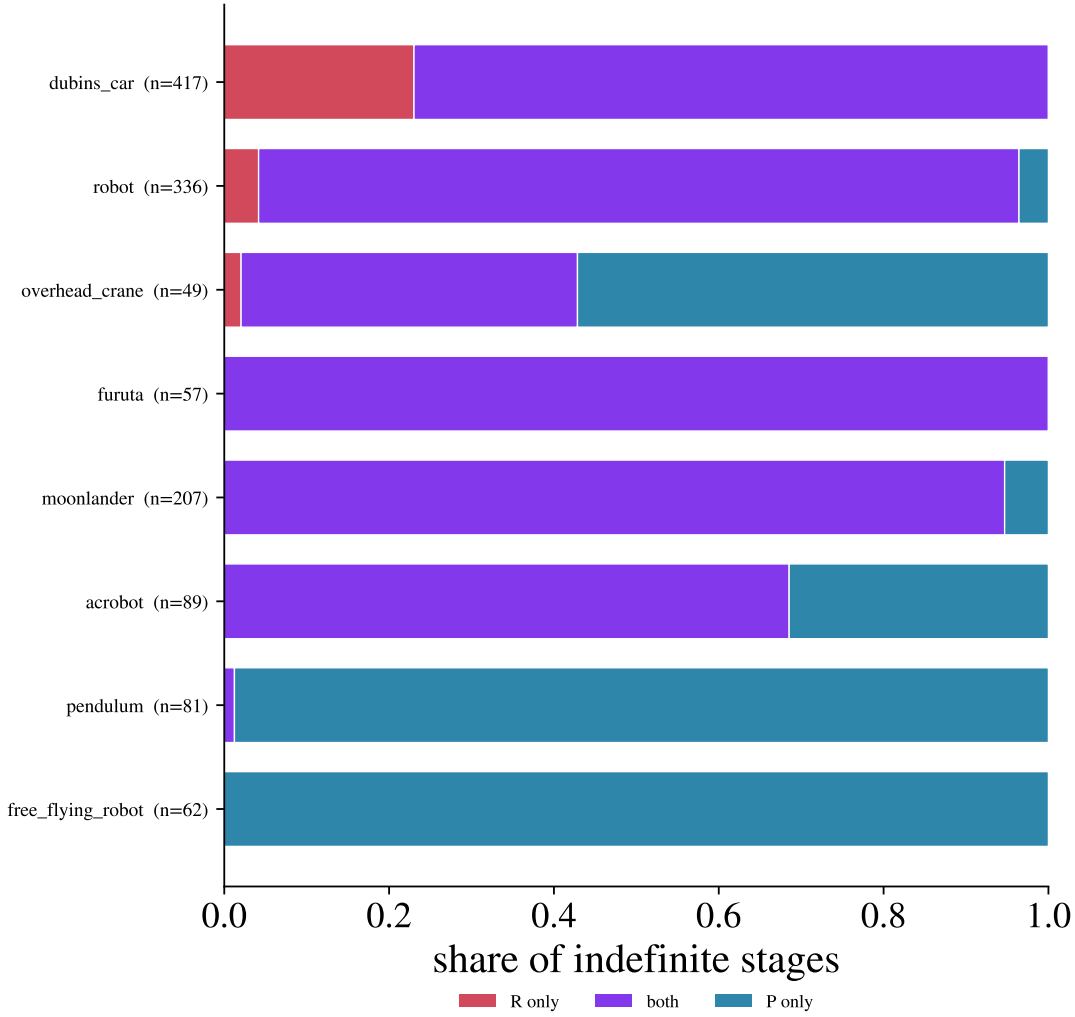


Figure 1.4: Per-problem composition of stage indefiniteness. Each bar shows, for one OCP, the share of stages that are R only, both, P only, (n is the number of collected indefinite stages).

While Figure 1.3 gives the aggregate picture, Figure 1.4 breaks the same classification down per problem and reveals a strongly problem-dependent and

clean structure: six of the eight OCPs are dominated by a single category ($\geq 90\%$ of their stages), and only `acrobot` and `overhead_crane` are genuinely mixed.

As observed from Figure 1.4, most of the problems are governed by the *both* case, including `dubins_car`, `robot`, `moonlander`, `acrobot`, `furuta`. Notably, three problems (`dubins_car`($n = 417$), `robot`($n = 336$) and `moonlander`($n = 207$)), contribute by far the most indefinite stages, they largely account for the aggregate dominance of the *both* case reported above.

The *P only* case dominates the remaining problems: `free_flying_robot` and `pendulum` are indefinite purely through the cost-to-go block, and `overhead_crane` leans the same way. Taken over all stages, the cost-to-go block is indefinite in the large majority of cases, whereas the *R only* case is marginal: it appears in only three problems and is negligible for `robot` and `overhead_crane`, with `dubins_car` being the sole problem where isolated *R*-block indefiniteness is substantial. Indefiniteness therefore predominantly involves the cost-to-go block, either alone or jointly with *R*, while *R*-block-only indefiniteness is the exception.

In conclusion, these figures numerically shows that, across the benchmark the cost-to-go block is involved in the vast majority of indefinite stages, a regularization strategy that merely enforces $M \succ 0$ would therefore leave the negative curvature unaddressed in most stages. Since the cost-to-go term enters the reduced Hessian through $M_k = R_k + B^\top P_{k+1} B$, keeping P_{k+1} (and hence M_k) definite is essential for a reliable regularization method, whereas treating the control cost block alone is not sufficient.

Bibliography

- [1] D. Kiessling, *Sweet-ocp: A test suite of direct optimal control problems*, version 0.1. 2026. [Online]. Available: https://github.com/david0oo/sweet_ocp.

